LINFO1115 June 2023

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TOTAL POINTS

15 / 31

QUESTION 1

Structural balance 5 pts

1.1 Definition of structural balance **0 / 2**

1.2 Structural balance theorem **0 / 3**

QUESTION 2

Second-price auction theorem 5 pts

2.1 Second-price auction in the language of game theory **0 / 2**

2.2 Second-price auction theorem **0 / 3**

QUESTION 3

Matching markets 7 pts

3.1 Market clearing **1 / 1**

3.2 Perfect matching **1 / 1**

3.3 Preferred-seller graph **2 / 2**

3.4 Social optimality **0 / 3**

QUESTION 4

Link farms 7 pts

4.1 Link farm definition **3 / 3**

4.2 Operation of a link farm **1 / 4**

QUESTION 5

Direct-benefit effects 7 pts

5.1 Market without direct-benefit effects **3 / 3**

5.2 Market with direct-benefit effects **4 / 4**

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LINFO1115 Final Exam

June 8, 2023

(Note: Q1 & Q2 taken together correspond to the midterm.)

Q1 Structural balance

Q1.1 Definition of structural balance [2pts]

Give a precise definition when a graph is *balanced*. Don't forget to say for *what graphs* balance is defined.

Q1.2 Structural balance theorem [3pts]

The balance theorem makes a connection between a local property and a global property for a graph. State the *balance theorem* and give a proof.

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Q2 Second-price auction theorem

Q2.1 Second-price auction in the language of game theory [2pts]

Assume each bidder i is a player and bidder i's strategy is the bid amount b_i as function of i's true value v_i. A game is played when each bidder bids once; one bidder wins and the others lose. In your answer, define the *payoff* to each bidder i.

Q2.2 Second-price auction theorem [3pts]

We proved the theorem that in a sealed second-price auction, it is a *dominant strategy* for each bidder to choose a bid b_i equal to the bidder's true value v_i. To prove the theorem, we assume that each bidder bids $b_i = v_i$ and we show that no deviation from this will improve the payoff. The proof considers two cases: (1) i raises bid, and (2) i lowers bid. For this question, choose one of these two cases and explain precisely why i's payoff will not improve in that case. You can make a diagram, as done in the course.

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Q3 Matching markets

Q3.1 Market clearing [1pt]
For any market with buyers and sellers, define the concept of *market clearing*.

For a given market with buyers and sellers, we say that it is market clearing if each buyer has bought a product and each product has been bought (i.e. there is no more offer and demand).

Q3.2 Perfect matching [1pt]
Define the concept of a *perfect matching* in a bipartite graph.

In a bipartite graph, we say that there is a perfect matching if each node is the assignee of exactly one node of the other part of the ~~node~~ graph and if each node is assigned to exactly one node of the other part of the graph (i.e. each node cannot be assigned to / the assignee of two other nodes).

Q3.3 Preferred-seller graph [2pts]
Given a collection of buyers and a collection of sellers in a bipartite matching market. Each buyer has a valuation for each item held by a seller. Define the *preferred-seller graph* for this market. As part of this definition, you should also define and use the *buyer payoff*.

~~De~~ The buyer payoff is defined as $V_{ij} - P_i$ (V_{ij} is the valuation that buyer j gives to good i, P_i is the price of good i).
The preffered seller graph is a buyer - seller bipartite graph where each buyer announces its preffered good (i.e. the one that maximizes its payoff, it can choose multiple goods for equal payoffs).

Q3.4 Social optimality [3pts]
We observe that a perfect matching in the preferred-seller graph is market clearing. For this question, prove that this perfect matching is *socially optimal*. Hint: your proof needs to reason about buyer and seller payoffs.

I don't remember the proof, however I do remember that it is proved by showing that the potential energy of the graph equals to 0. (this, basically, is shown by taking the sum of buyer and seller payoffs and showing that no energy is lost, therefore it is socially optimal).

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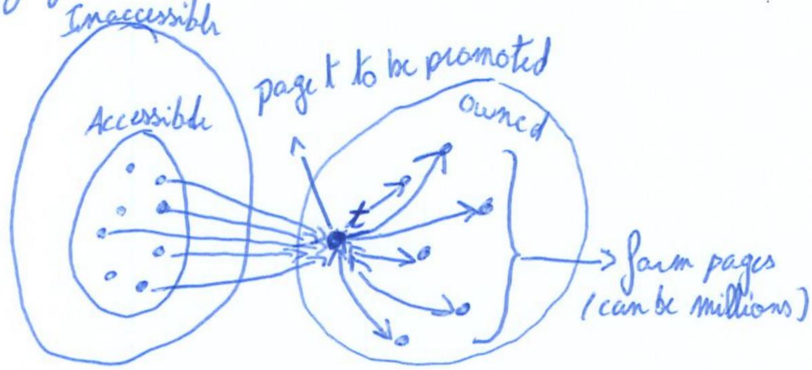
Q4 Link farms

Q4.1 Link farm definition [3pts]

Explain what is a *link farm* and give the *graph structure* of a typical link farm.

A link farm is a web structure designed to artificially boost the Page Rank score of a given webpage t. It uses farm pages to inflate t's score by establishing bi-directionnal links between t and the farm pages. It also uses links from unowned pages to t (blog/forum posts, social media, ...).



Q4.2 Operation of a link farm [4pts]

Using the random walk definition of Page Rank, explain the *operation* of a link farm. How does the link farm *achieve its effect*? Could you set up a company to provide link farm services to third parties?

Let's start our random walk on a webpage on the accessible web (assuming page t, the page to be promoted, can be reached). As long as the random walk does not go through a node linking to t, the link farm has no effect whatsoever. When it goes through a node linking to t, if the walker decides to transition to t, then the trap is set in motion. The walker is from now on stuck in the link farm since there is no link leading to the outside. Effectively, the walker gets stuck in a never-ending cycle between t and the ~~link~~ farm pages (1 out of 2 pages he visits is t, the other one is a farm page. This artificially inflates the score of t. Link farm don't need t to be the best page (in terms of Page Rank score), they just need to be better than other similar pages (e.g. "George Bush" → "Miserable failure"). While in theory, the simple Page Rank model shows that this is possible and feasible, such a business endeavour would most likely lead to bankruptcy as modern search engines have implemented numerous mitigations against link farms (evaporation, Trust Rank, etc.).

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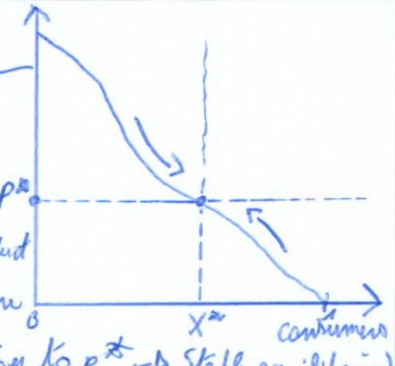
Q5 Direct-benefit effects

Q5.1 Market without direct-benefit effects [3pts]

In the course we defined a simple continuous model of a market without direct-benefit effects. Consumers are given names x as real numbers from 0 to 1, uniformly distributed from 0 to 1. Each consumer has a reservation price r(x), arranged in decreasing order. Using a graph, explain the relationship between the price of a product and the consumers who buy the product. Explain why the production cost p* defines a *stable equilibrium* in this market with equilibrium price x*. Why is it stable?

~~First~~ First, the price is at least P^* (or the company would lose money). Furthermore, it is an equilibrium because customers with $X < X^*$ want to buy the product (they make a surplus on it). Customers with $X > X^*$ do not want to buy the product (they lose from ~~by~~ buying it). Hence there is downward and upward pressure leading to p^* → Stable equilibrium. Finally it is ~~stab~~ stable because the area under the curve given p^* is maximized with this equilibrium.

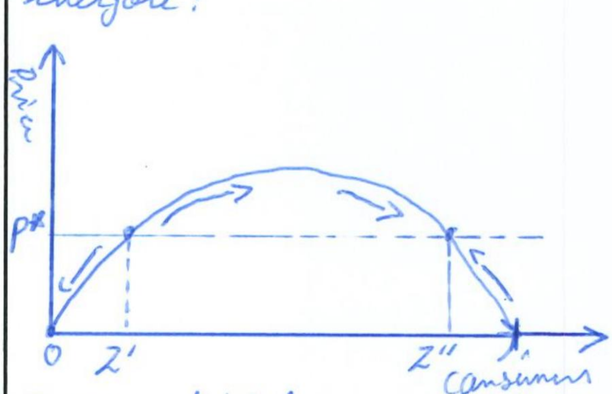


Q5.2 Market with direct-benefit effects [4pts]

Extend the model of the previous question to add direct-benefit effects. Add a second function f(z) that quantifies the benefit to each consumer when a fraction z of all consumers has bought the product.

- What is the formula for the *new equilibrium price* in this extended market?
- Make simple assumptions about the form of the functions r(x) and f(z), as we did in the course. Given these assumptions, what is the *new graph* of consumers versus price?
- What are the *equilibria* in this new graph?
- Explain for each equilibrium whether it is *stable* or *unstable* and justify your explanations.

The formula for the new equilibrium price is $r(X) f(z)$. It can be assumed that $r(X) = 1 - x$ and $f(z) = z$, the new graph is therefore:



There are three equilibria: $z = 0$, $z = z'$ and $z = z''$.
$z = 0$ is stable because when $z < z'$, there are customers that bought the product but didn't want to (negative payout), there is downward pressure to buy. (see arrows on graph).

$z = z''$ is stable because with ~~z>z'~~ $z'' > z > z'$, there are customers that ~~bough~~ haven't bought the product that want to (upward pressure to buy). With $z > z''$, there are customers that bought the product but didn't want to (downward pressure). (see arrows on graph). From what I just explained, it can be deduced that $z = z'$ is an unstable equilibrium